

DATE CAM-monitoring paper – what to focus on

For Indranil, from MN, Sep. 2. I am tied up with other projects for the next few days, so please drive the items below. The current draft is v07, in the shared Drive folder next to v06; the “With Comments” PDFs show the open flags in orange and violet. The items are in priority order. If something is unclear, flag it in the .tex or send me a note – do not wait on me.

1. Matched cell-count experiment (do this first): Right now the paper compares the ternary and thermometer codes at the same number of *bits* (384), and ternary wins cross-domain (average 0.640 vs. 0.579; 63 of 90 cells). That said, a ternary cell may cost about twice the hardware of a binary cell, so the fair comparison gives thermometer twice the code length. Re-run the ten-domain grid with the thermometer code at roughly 768 bits, same splits. A reviewer will ask for this, and the result decides which encoding we lead with in Sec. V-C. *Send back:* the diagonal and off-diagonal means, and the cell-by-cell win count against ternary at 384 b.

2. Combined-training score: The source paper’s strongest result for keeping the library small is its “combined” row – one probe trained on all of their datasets pooled, which scores well everywhere (0.97 even on sycophancy). Our grids have no combined row. Build a bank from all ten domains pooled and score it on each of the ten test domains, for both 384-b encodings and the full-precision ceiling. This matters because the combined row is the best case *against* our large- K argument, and we should report our own numbers for it rather than argue around theirs (see objection (1) in the appendix document). *Send back:* one ten-value AUROC row per encoding, plus the ceiling’s.

3. Fig. 2(d)–(f) data and names: Confirm the data behind panels (d)–(f) is the current run – this slipped once before. Also check our domain names against the source paper: they call their five truth types definitional, empirical, fictional, logical, and ethical; our grids say “evidential” and “ethics.” If ours are the same datasets, relabel to match; if not, tell me what they actually are. Same check for roleplay vs. roleplaying and insider_trd vs. insider trading.

4. Ternary dead zone: Our “ternary” code maps small projection values to don’t-care. Confirm the dead zone was actually turned on in the 384-b runs. If it was off, the code was binary in practice, and the cell-type claims in Secs. IV and V and Table II need to change.

5. Audit at 384 b: The destroyed-vector audit numbers in Sec. V-E (mean 0.50; 83% of cells between 0.4 and 0.6) come from the 2,048-b grid. Run, or locate, the same audit at 384 b. If we only have it at 2,048 b, say so and the text will state that instead.

6. Which four-domain numbers are current: The paper quotes the undergrads’ deck (average gap -0.03 , worst cell -0.09); the Aug. 14 methods document said -0.06 / within 0.05. Tell me which run is current, so the paper quotes one set of numbers.

7. The decision rule – walk me through it: Sec. IV-E says we store a few thousand labeled example activations per domain and classify a query by a majority vote of its 31 nearest stored rows. My note on the draft flagged that I am not sure this matches what the runs actually do. Walk me through what is stored and what is compared. If it is not thousands of exemplars with a $k=31$ vote, then that subsection, the audit discussion, and Table II’s bank sizes all need to change.

8. If time remains: (a) The recombination experiment: a frozen random projection, quantized to 3 bits, matched on a Euclidean matchline, at matched bank size on the same grid. A few days of work, and it settles whether the Euclidean and Hamming tracks are two paths or one. (b) A tuned GPU baseline (GEMV/FAISS) at batch 1 and batch 64, for Sec. VII’s estimates.

Items 1–6 check or extend data you already have; item 7 is an explanation I need from you; item 8 is new runs. Leave results where the paper can cite them – .npz files next to the existing ones, plus your deck or a short note – and mark anything you change in the .tex with your `\ib{}` flag so I can find it.